

Scenario-Based Questions: Undoing Changes in Git

These real-time scenarios are designed to help students practice Git undo operations and prepare for interview questions. Try to solve each scenario using appropriate Git commands and explain your reasoning.

Scenario 1: Undo Unstaged Changes

- You modified a file `main.py` but have not staged it yet.
- After reviewing the changes, you realize they are incorrect and want to discard them completely.
- Questions:
 - • How will you undo the changes?
 - • Which command restores the file to the last committed state?

Scenario 2: Undo Staged Changes

- You edited `index.html` and staged it using `git add`.
- Before committing, you realize the changes should not be part of the next commit.
- Questions:
 - • How do you unstage the file without losing the changes?
 - • Which Git command helps here?

Scenario 3: Modify Last Commit Message

- You committed changes but made a mistake in the commit message.
- The commit is not yet pushed to the remote repository.
- Questions:
 - • How will you correct the commit message?
 - • Will this create a new commit? Explain.

Scenario 4: Add Missed File to Last Commit

- You committed your changes but forgot to include `config.yaml`.
- The commit has not been pushed yet.
- Questions:
 - • How will you add the missed file to the previous commit?
 - • Which command helps avoid creating a new commit?

Scenario 5: Undo Last Commit (Keep Changes)

- You made a commit, but after testing, you realize the logic needs improvement.
- You want to undo the commit but keep the code changes for further editing.
- Questions:
 - • Which reset mode will you use?
 - • What happens to the working directory?

Scenario 6: Undo Last Commit (Discard Changes)

- You committed a feature that is completely wrong.
- You want to remove the commit and discard all related changes permanently.
- Questions:
 - • Which Git command should you use?
 - • What is the risk involved in this action?

Scenario 7: Revert a Commit Already Pushed

- You pushed a commit to the remote repository.
- Later, you realize the commit introduced a bug.
- Questions:
 - • Why should you not use git reset here?
 - • Which command safely undoes the commit?

Scenario 8: Undo Changes in Multiple Files

- You edited multiple files in your working directory.
- You want to discard changes only in one specific file.
- Questions:
 - • How will you restore only that file?
 - • What command ensures other file changes remain intact?

Scenario 9: Accidental git add .

- You accidentally ran git add . and staged many unwanted files.
- You want to unstage everything but keep the changes.
- Questions:
 - • Which command will you use?
 - • How can you verify the result?

Scenario 10: Undo a Commit from Two Commits Ago

- You want to undo a commit made two commits earlier.
- The branch is shared with your team and already pushed.
- Questions:
 - • Which approach is safest?
 - • How does Git maintain history in this case?